Deploying a Scalable Web Application Using AWS Elastic Beanstalk

📘 Scenario Description:

A startup company aims to simplify application deployment and management while ensuring scalability and reliability. To achieve this, the team will leverage AWS Elastic Beanstalk, which abstracts the complexity of infrastructure provisioning, application deployment, and auto scaling.

Scenario Steps:

Setting Up the Beanstalk Environment:

1- Create a Web Application Package
Develop a simple web application using Node.js, Python (Flask), or PHP.
Package the code and dependencies in a .zip file as per AWS Beanstalk requirements.

2- Deploy the Application using Elastic Beanstalk Console
Create a new Elastic Beanstalk environment (Web server environment).
Choose a platform (e.g., Node.js, Python, PHP).
Upload the application package and deploy.

3- Enable Auto Scaling
Use the Elastic Beanstalk configuration options to enable automatic scaling based on CPU or request count.
Configure minimum and maximum instance counts.

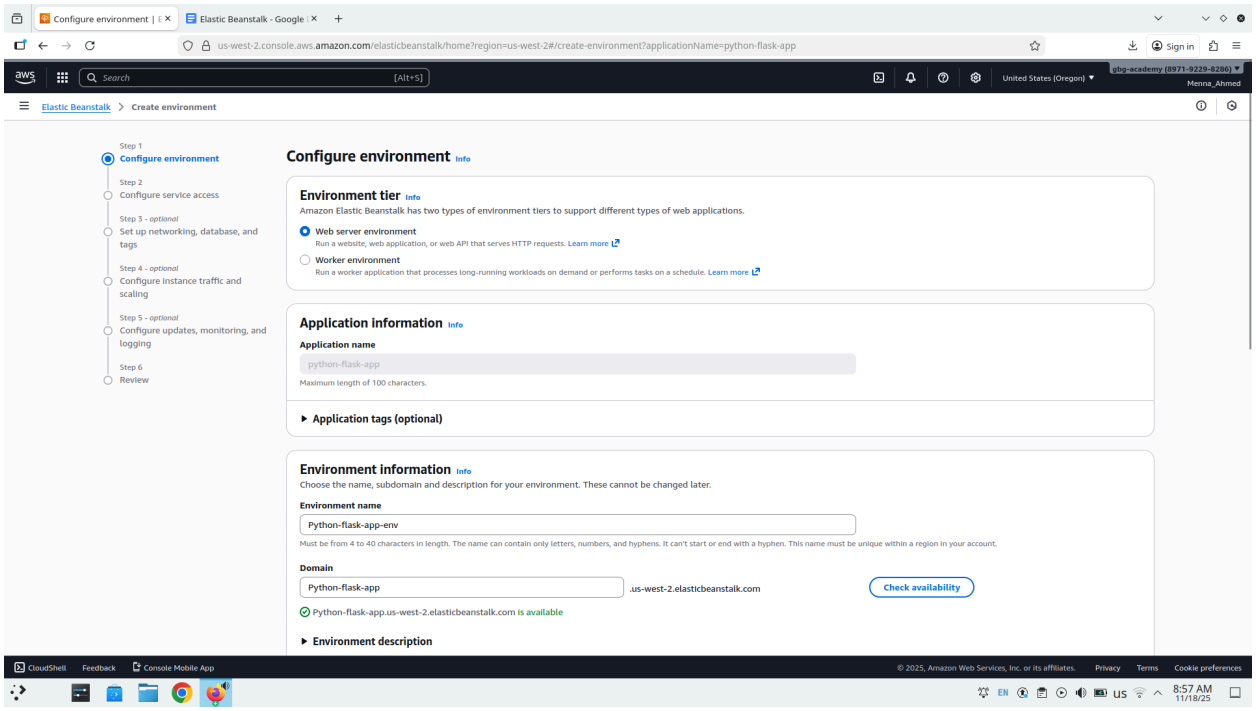
4- Attach Load Balancer
Elastic Beanstalk automatically creates and configures an Application Load Balancer (ALB).
Validate health checks and traffic routing.

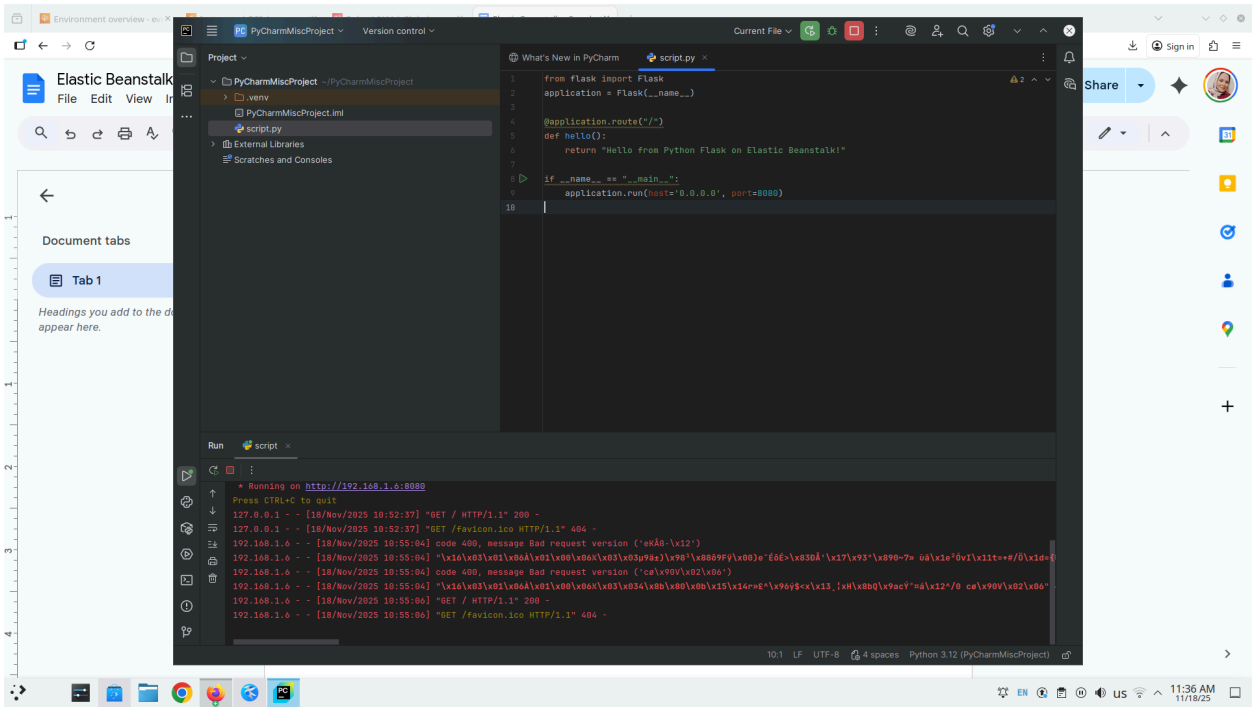
Scenario Testing:

Run Application Functionality Tests

Go to console and let’s start:-

create Elastic BeanStalk





Configure environment | E × Elastic Beanstalk - Google E ×

us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app

Sign in

aws Search [Alt+S] United States (Oregon) gbg-academy (8971-9229-8286) Menna_Ahmed

Elastic Beanstalk > Create environment

Python-flask-app .us-west-2.elasticbeanstalk.com Check availability

Python-flask-app.us-west-2.elasticbeanstalk.com is available

▸ Environment description

Platform Info

Platform: Python

Platform branch: Python 3.11 running on 64bit Amazon Linux 2023

Platform version: 4.7.5 (Recommended)

Application code Info

- ○ Sample application
- ○ Existing version
  Application versions that you have uploaded.
- ◉ Upload your code
  Upload a source bundle from your computer or copy one from Amazon S3.

Version label

Unique name for this version of your application code.

firsttestV1

Source code origin. Maximum size 500 MB

- ◉ Local file

Upload application

Choose file

File name: **flask_eb_app.zip**

File must be less than 500MB max file size

- ○ Public S3 URL

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File must be less than 500MB max file size

- ○ Public S3 URL

Presets Info

Start from a preset that matches your use case or choose custom configuration to unset recommended values and use the service's default values.

Configuration presets

- ◉ Single instance (free tier eligible)
- ○ Single instance (using spot instance)
- ○ High availability
- ○ High availability (using spot and on-demand instances)
- ○ Custom configuration

Cancel Next

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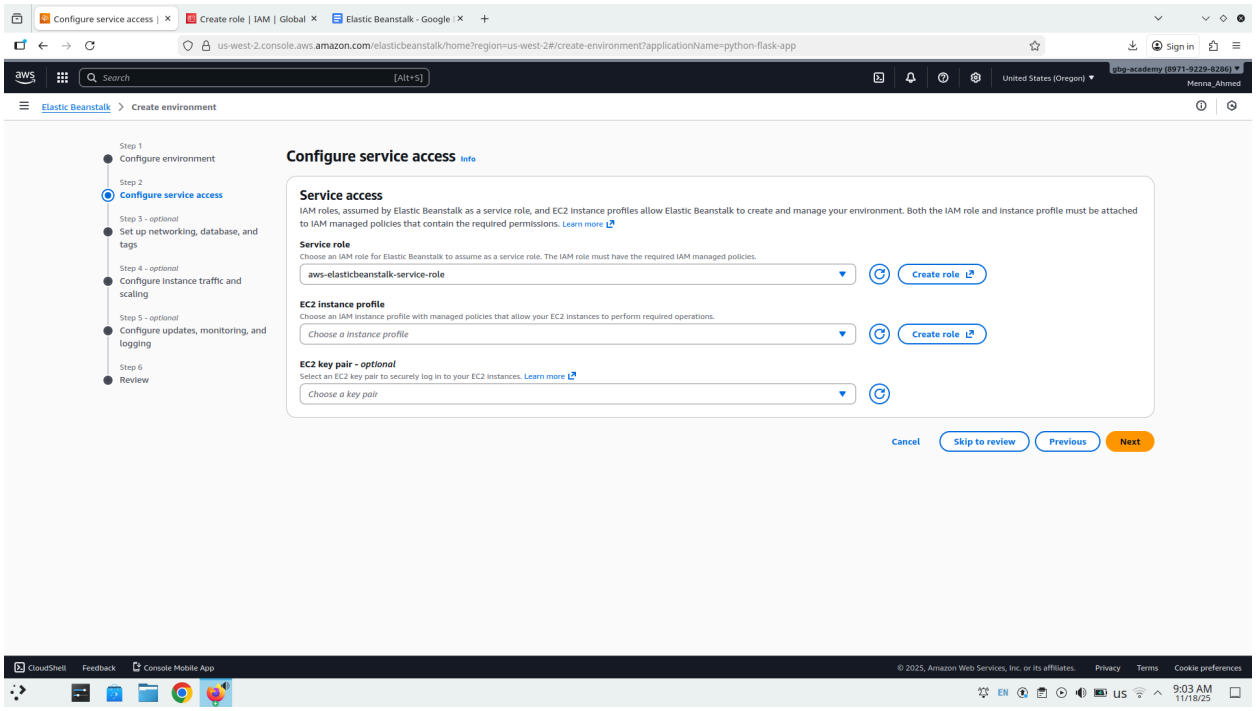
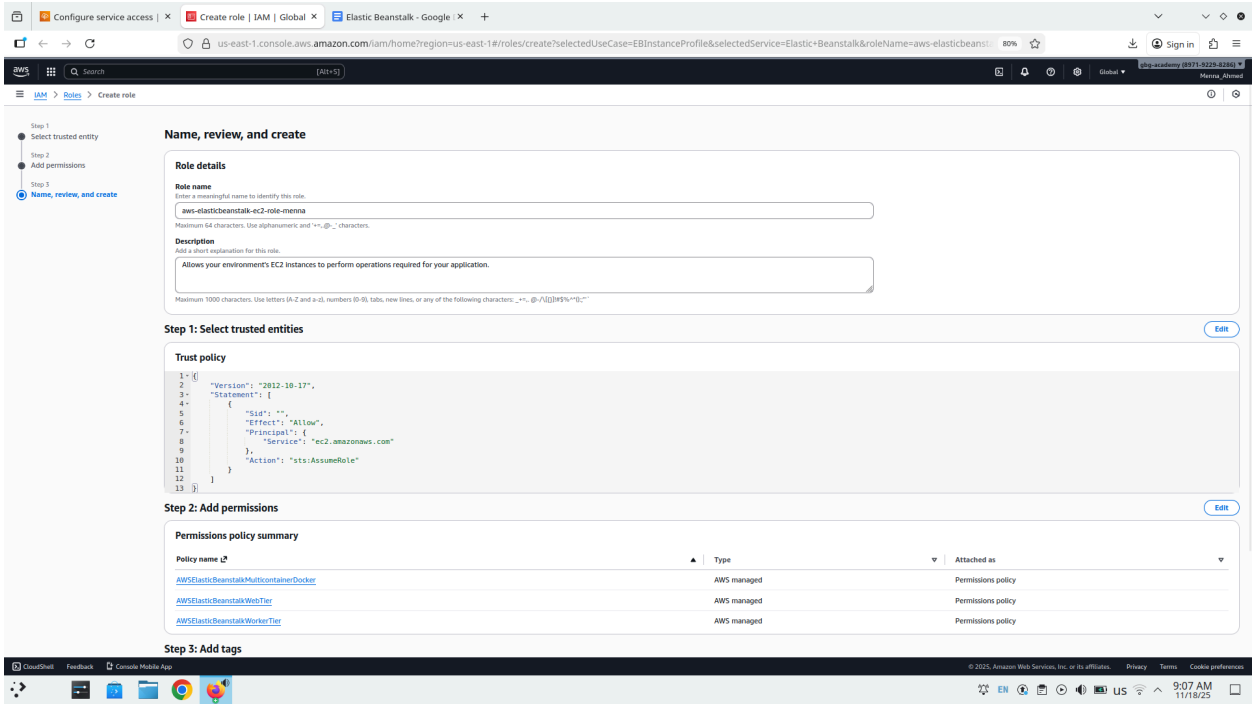
Next



I will create IAM Role and give role to Elastic beanstalk



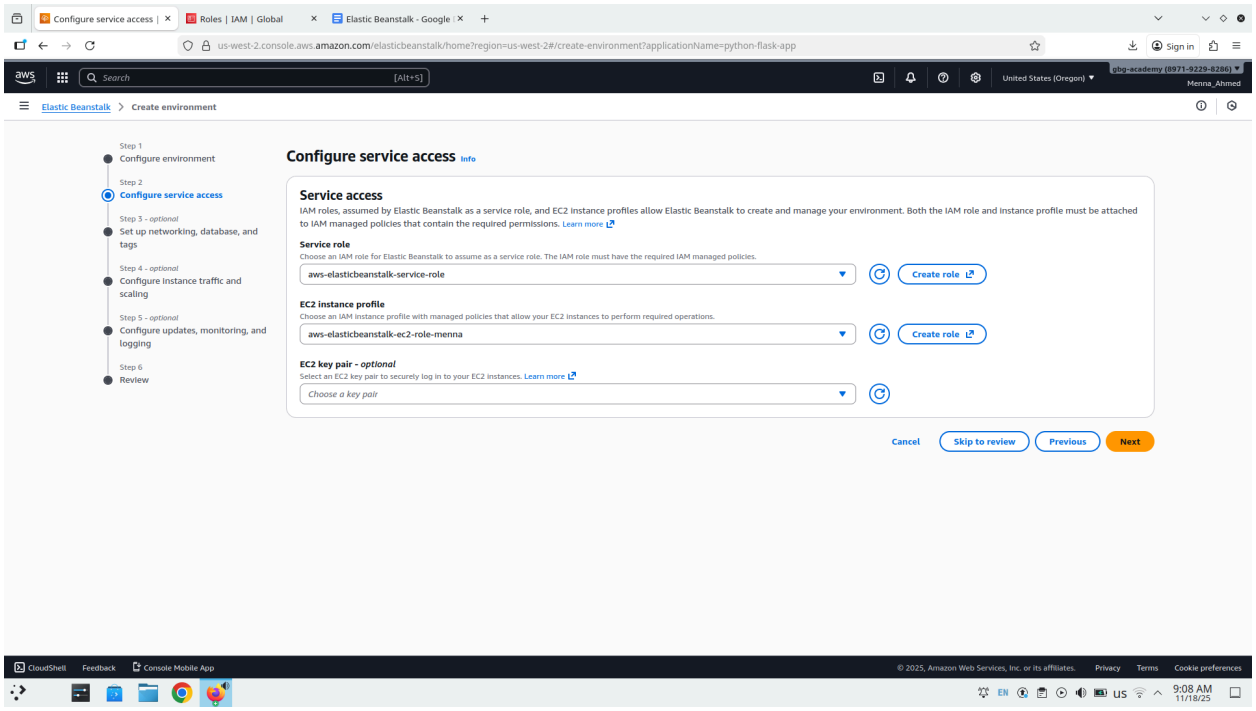
3 AWS managed policies are the required default policies for the EC2 instance in Elastic Beanstalk:

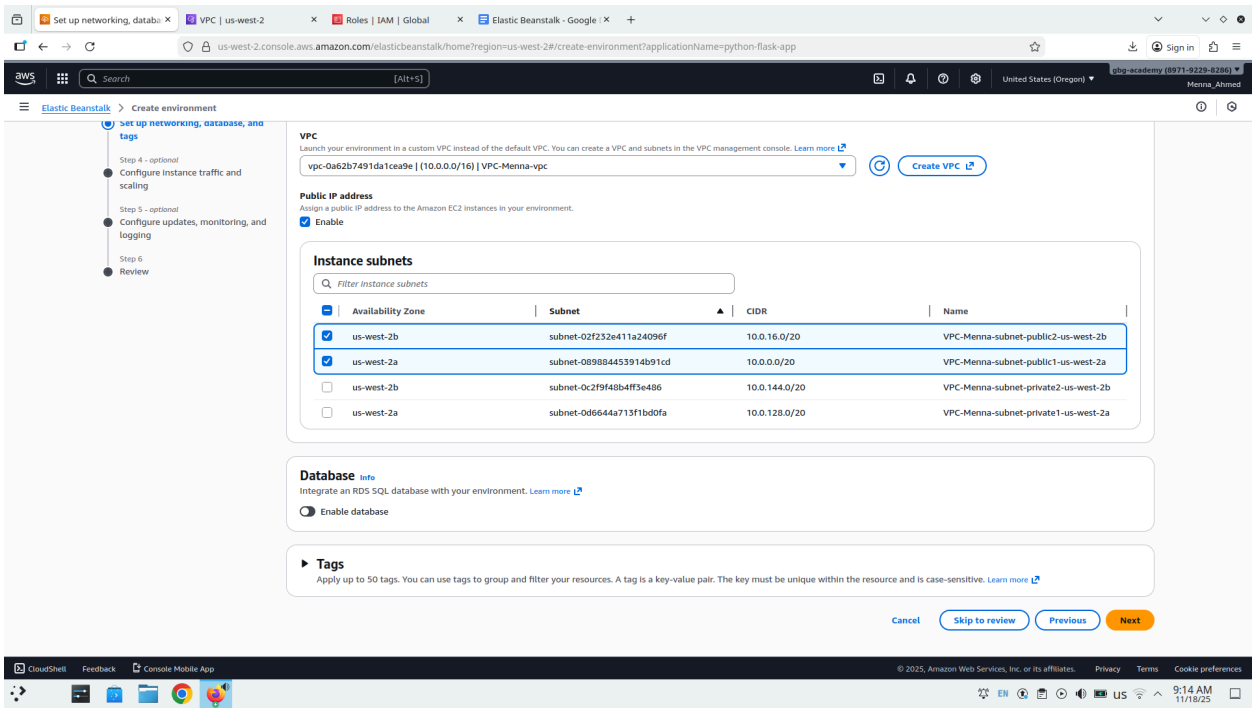
- AWSElasticBeanstalkMulticontainerDocker
- AWSElasticBeanstalkWebTier
- AWSElasticBeanstalkWorkerTier

These policies give the EC2 instance the right permissions to:

- Download your application from S3
- Write logs to CloudWatch
- Communicate status to Elastic Beanstalk
- Run your web app container
- Handle updates and health checks

Back to continue processing





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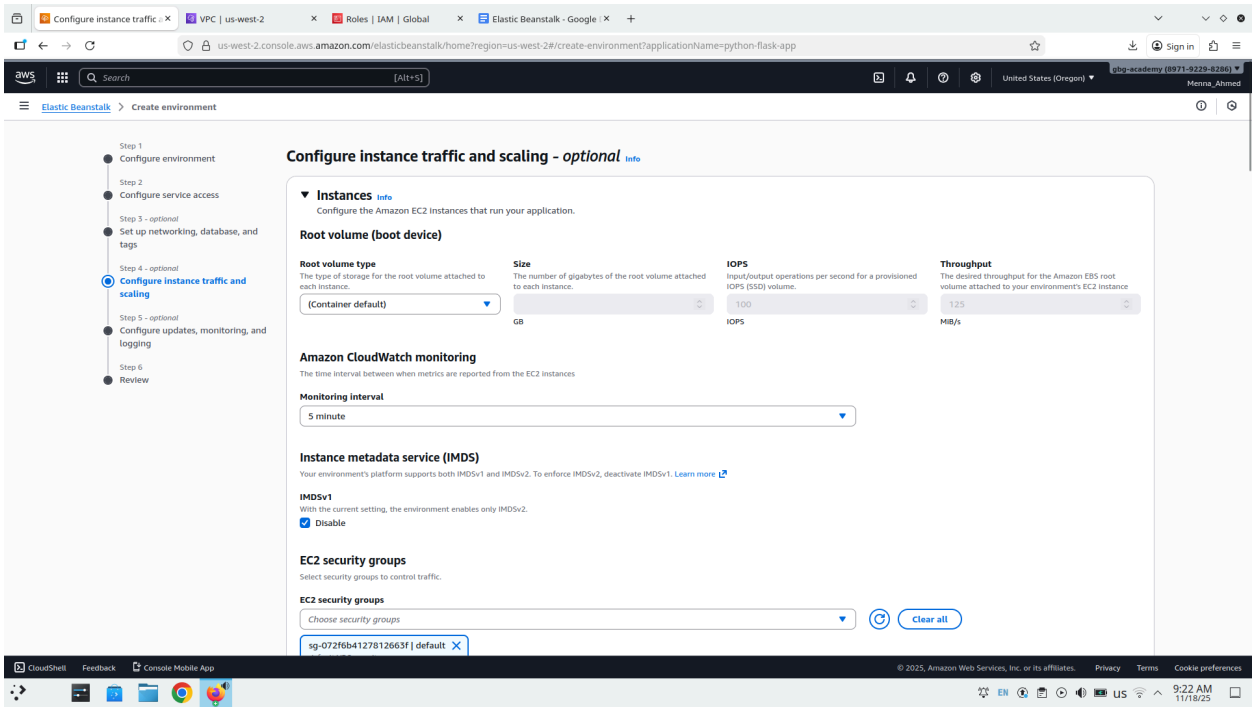
Configure instance traffic
VPC | us-west-2
Roles | IAM | Global
Elastic Beanstalk - Google
us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app
Elastic Beanstalk > Create environment
Step 1 Configure environment
Step 2 Configure service access
Step 3 - optional Set up networking, database, and tags
Step 4 - optional Configure instance traffic and scaling
Step 5 - optional Configure updates, monitoring, and logging
Step 6 Review
Configure instance traffic and scaling - optional Info
Instances Info
Configure the Amazon EC2 instances that run your application.
Root volume (boot device)
Root volume type
The type of storage for the root volume attached to each instance.
(Container default)
Size
The number of gigabytes of the root volume attached to each instance.
GB
IOPS
Input/output operations per second for a provisioned IOPS (SSD) volume.
100
IOPS
Throughput
The desired throughput for the Amazon EBS root volume attached to your environment's EC2 instance
125
MiB/s
Amazon CloudWatch monitoring
The time interval between when metrics are reported from the EC2 instances
Monitoring interval
5 minute
Instance metadata service (IMDS)
Your environment's platform supports both IMDSv1 and IMDSv2. To enforce IMDSv2, deactivate IMDSv1. Learn more
IMDSv1
With the current setting, the environment enables only IMDSv2.
Disable
EC2 security groups
Select security groups to control traffic.
EC2 security groups
Choose security groups
Clear all
sg-072f6b4127812663f | default
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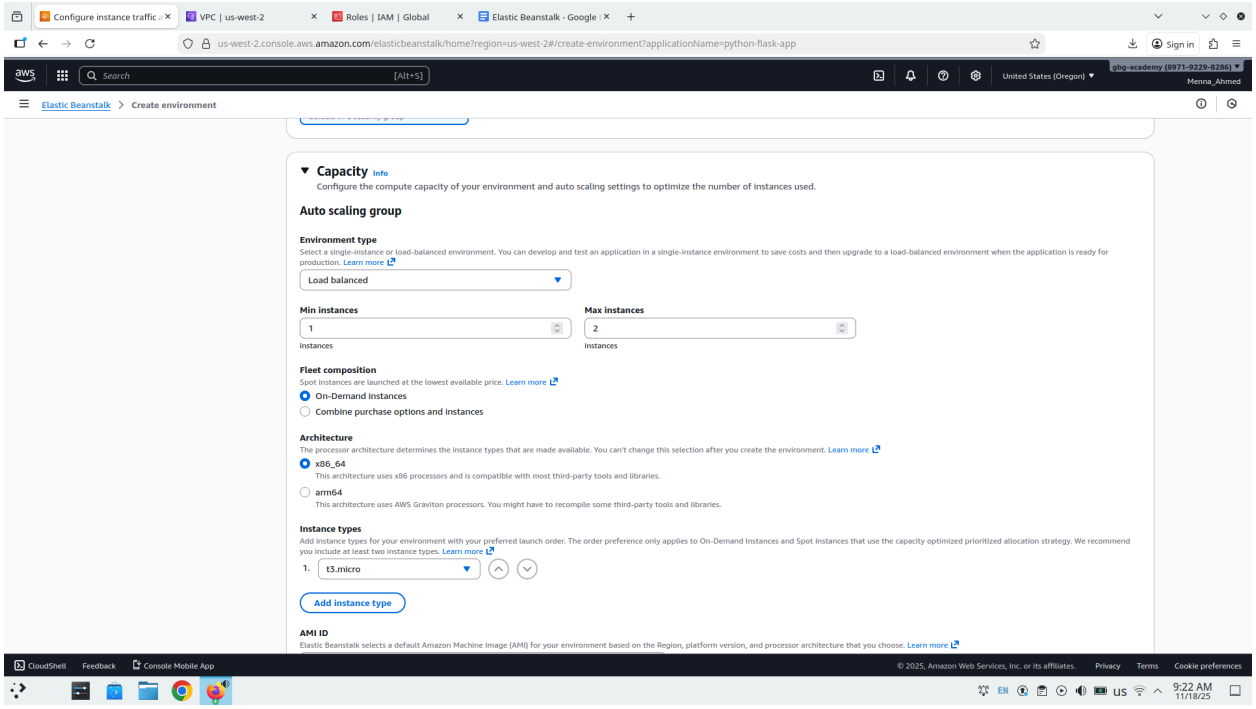
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VPC | us-west-2
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Elastic Beanstalk - Google
us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app
Elastic Beanstalk > Create environment
Capacity Info
Configure the compute capacity of your environment and auto scaling settings to optimize the number of instances used.
Auto scaling group
Environment type
Select a single-instance or load-balanced environment. You can develop and test an application in a single-instance environment to save costs and then upgrade to a load-balanced environment when the application is ready for production. Learn more
Load balanced
Min instances
1
instances
Max instances
2
instances
Fleet composition
Spot instances are launched at the lowest available price. Learn more
On-Demand instances
Combine purchase options and instances
Architecture
The processor architecture determines the instance types that are made available. You can't change this selection after you create the environment. Learn more
x86_64
This architecture uses x86 processors and is compatible with most third-party tools and libraries.
arm64
This architecture uses AWS Graviton processors. You might have to recompile some third-party tools and libraries.
Instance types
Add instance types for your environment with your preferred launch order. The order preference only applies to On-Demand Instances and Spot Instances that use the capacity optimized prioritized allocation strategy. We recommend you include at least two instance types. Learn more
1. t3.micro
Add instance type
AMI ID
Elastic Beanstalk selects a default Amazon Machine Image (AMI) for your environment based on the Region, platform version, and processor architecture that you choose. Learn more
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us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app

Elastic Beanstalk > Create environment

Elastic Beanstalk selects a default Amazon Machine Image (AMI) for your environment based on the Region, platform version, and processor architecture that you choose. Learn more

ami-095ca22cd8545b80a

Availability Zones
Number of Availability Zones (AZs) to use.
Any

Placement
Specify Availability Zones (AZs) to use.
Choose Availability Zones (AZs)

Scaling cooldown
360 seconds

Scaling triggers

Metric
Change the metric that is monitored to determine if the environment's capacity is too low or too high.
CPUUtilization

Statistic
Choose how the metric is interpreted.
Average

Unit
Bytes

Period
The period between metric evaluations.
5 Min

Breach duration
The amount of time a metric can exceed a threshold before triggering a scaling operation.
5 Min

Configure instance traffic | VPC | us-west-2 | Roles | IAM | Global | Elastic Beanstalk - Google

us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app

Elastic Beanstalk > Create environment

Application load balancer
Application layer load balancer—routing HTTP and HTTPS traffic based on protocol, port, and route to environment processes.

Classic load balancer
Previous generation — HTTP, HTTPS, and TCP

Network load balancer
Ultra-high performance and static IP addresses for your application.

Dedicated
Use a load balancer that Elastic Beanstalk creates exclusively for this environment.

Shared
Use a load balancer that someone in your account created. It can be shared among multiple Elastic Beanstalk environments.

Listeners

You can specify listeners for your load balancer. Each listener routes incoming client traffic on a specified port using a specified protocol to your environment processes. By default, we've configured your load balancer with a standard web server on port 80.

Actions | Add listener

Listener Port	Listener Protocol	SSL certificate	Default process	Enabled
80	HTTP	—	default	

Processes

For each environment process, you can specify the protocol and port that the load balancer uses to route requests to the process. You can also specify how the load balancer performs process health checks.

Actions | Add process

Name	Port	Protocol	HTTP code	Health check path	Stickiness
default	80	HTTP		/	Disabled

Configure updates, monit: | VPC | us-west-2 | Roles | IAM | Global | Elastic Beanstalk - Google | Google

us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app

Elastic Beanstalk > Create environment

Step 1
Configure environment

Step 2
Configure service access

Step 3 - *optional*
Set up networking, database, and tags

Step 4 - *optional*
Configure instance traffic and scaling

Step 5 - *optional*
Configure updates, monitoring, and logging

Step 6
Review

Configure updates, monitoring, and logging - *optional* Info

Monitoring Info

Health reporting

Enhanced health reporting provides free real-time application and operating system monitoring of the instances and other resources in your environment. The **EnvironmentHealth** custom metric is provided free with enhanced health reporting. Additional charges apply for each custom metric. For more information, see Amazon CloudWatch Pricing

System

- (•) Basic
- ( ) Enhanced

Health event streaming to CloudWatch Logs

Configure Elastic Beanstalk to stream environment health events to CloudWatch Logs. You can set the retention up to a maximum of ten years and configure Elastic Beanstalk to delete the logs when you terminate your environment.

Log streaming

standard CloudWatch charges apply.

[ ] Enable

Managed platform updates Info

Activate managed platform updates to apply platform updates automatically during a weekly maintenance window that you choose. Your application stays available during the update process.

Managed updates

[ ] Enable

Weekly update window

Thursday at 03 : 59 UTC

Update level

Minor and patch

Instance replacement

If enabled, an instance replacement will be scheduled if no other updates are available.

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us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app

Elastic Beanstalk > Create environment

Email notifications Info

Enter an email address to receive email notifications for important events from your environment. Learn more

Email

mennaahmed0640@gmail.com

Rolling updates and deployments Info

Application deployments

Choose how Amazon Elastic Beanstalk propogates source code changes and software configuration updates. Learn more

Deployment policy

All at once

Batch size type

- (•) Percentage
- ( ) Fixed

Deployment batch size

100

% instances at a time

Configuration updates

Changes to virtual machine settings and VPC configuration trigger rolling updates to replace the instances in your environment without downtime. Learn more

Rolling update type

Disabled

Deployment preferences

Customize health check requirements and deployment timeouts.

Ignore health check

Don't fail deployments due to health check failures.

False

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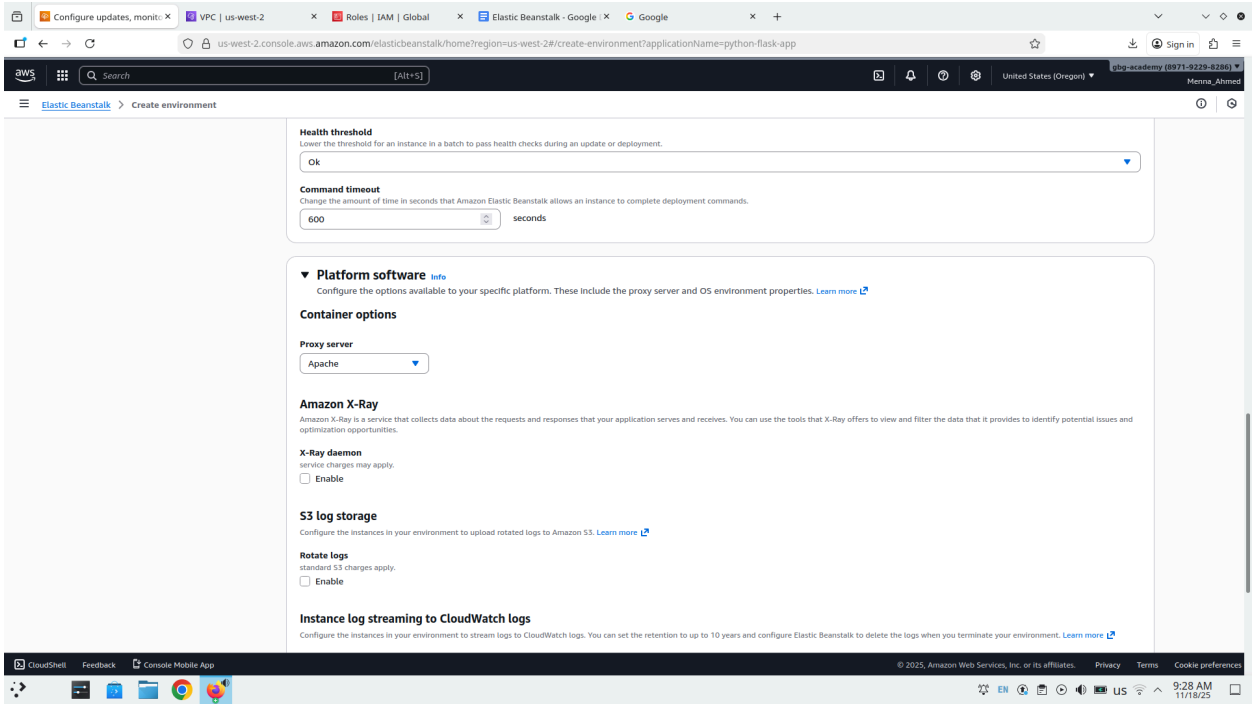
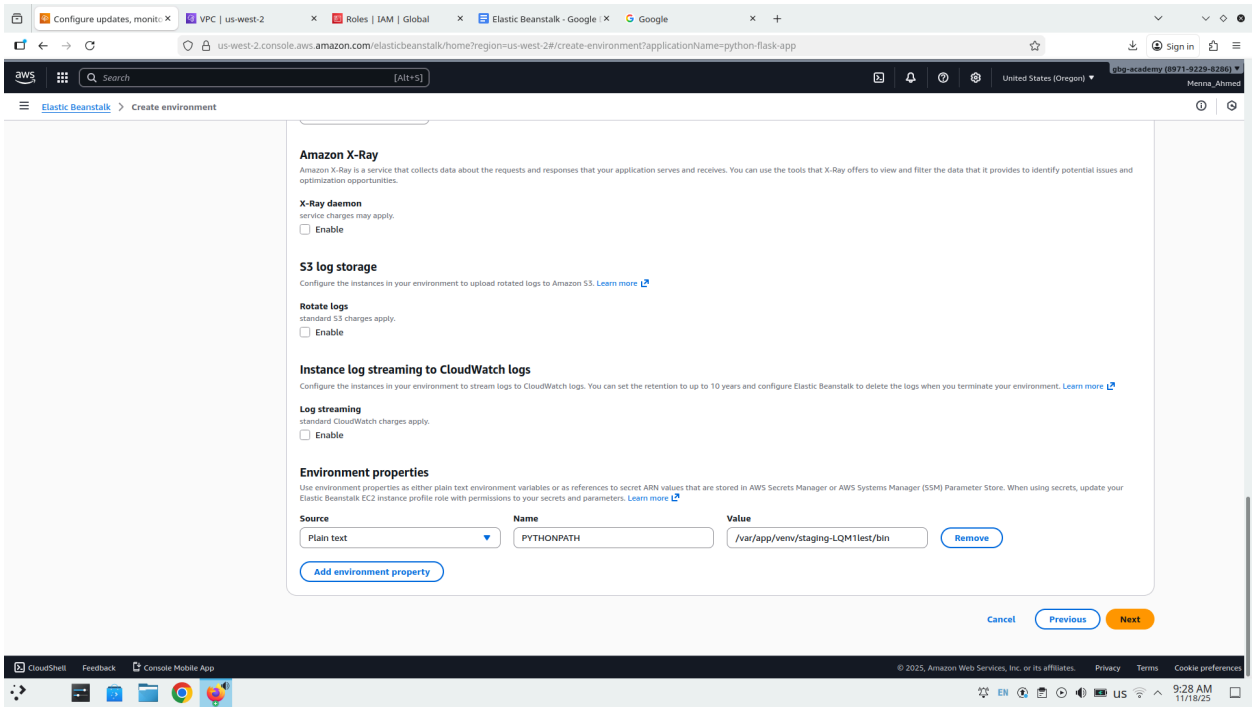

Elastic Beanstalk > Create environment
Health threshold
Lower the threshold for an instance in a batch to pass health checks during an update or deployment.
Ok
Command timeout
Change the amount of time in seconds that Amazon Elastic Beanstalk allows an instance to complete deployment commands.
600
seconds
Platform software Info
Configure the options available to your specific platform. These include the proxy server and OS environment properties. Learn more
Container options
Proxy server
Apache
Amazon X-Ray
Amazon X-Ray is a service that collects data about the requests and responses that your application serves and receives. You can use the tools that X-Ray offers to view and filter the data that it provides to identify potential issues and optimization opportunities.
X-Ray daemon
service charges may apply.
Enable
S3 log storage
Configure the instances in your environment to upload rotated logs to Amazon S3. Learn more
Rotate logs
standard S3 charges apply.
Enable
Instance log streaming to CloudWatch logs
Configure the instances in your environment to stream logs to CloudWatch logs. You can set the retention to up to 10 years and configure Elastic Beanstalk to delete the logs when you terminate your environment. Learn more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Configure the instances in your environment to stream logs to CloudWatch logs. You can set the retention to up to 10 years and configure Elastic Beanstalk to delete the logs when you terminate your environment. Learn more
Log streaming
standard CloudWatch charges apply.
Enable
Environment properties
Use environment properties as either plain text environment variables or as references to secret ARN values that are stored in AWS Secrets Manager or AWS Systems Manager (SSM) Parameter Store. When using secrets, update your Elastic Beanstalk EC2 instance profile role with permissions to your secrets and parameters. Learn more
Source
Name
Value
Plain text
PYTHONPATH
/var/app/venv/staging-LQM1est/bin
Remove
Add environment property
Cancel
Previous
Next

Then next

Overview

Configure environment - n | VPC | us-west-2 | Roles | IAM | Global | Elastic Beanstalk - Google | Google

us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app

Elastic Beanstalk > Create environment

Step 1 Configure environment
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Step 6 Review

Review Info

Step 1: Configure environment

Edit

Environment information

Environment tier
Web server environment

Application name
python-flask-app

Environment name
Python-flask-app-env

Application code
flask_eb_app.zip

Platform
arn:aws:elasticbeanstalk:us-west-2::platform/Python 3.11 running on 64bit Amazon Linux 2023/4.7.5

Step 2: Configure service access

Edit

Service access Info
Configure the service role and EC2 instance profile that Elastic Beanstalk uses to manage your environment. Choose an EC2 key pair to securely log in to your EC2 instances.

Service role
arn:aws:iam::897192298286:role/aws-elasticbeanstalk-service-role

EC2 instance profile
aws-elasticbeanstalk-ec2-role-menna

Step 3: Set up networking, database, and tags

Edit

Networking, database, and tags Info
Configure VPC settings, and subnets for your environment's EC2 instances and load balancer. Set up an Amazon RDS database that's integrated with your environment.

Network

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Elastic Beanstalk > Create environment

Step 4: Configure instance traffic and scaling

Edit

Instance traffic and scaling Info
Customize the capacity and scaling for your environment's instances. Select security groups to control instance traffic. Configure the software that runs on your environment's instances by setting platform-specific options.

Instances

IMDSv1
Disabled

EC2 Security Groups
sg-072f6b412781266 3f

Capacity

Environment type
Load balanced

Min instances
1

Max instances
2

Fleet composition
On-Demand instances

On-demand base
0

On-demand above base
70

Capacity rebalancing
Disabled

Scaling cooldown
360

Processor type
x86_64

Instance types
t3.micro

AMI ID
ami-095ca22cd8545b80a

Availability Zones
Any

Metric
CPUUtilization

Statistic
Average

Unit
Bytes

Period
5

Breach duration
5

Upper threshold
6000000

Scale up increment
1

Lower threshold
2000000

Scale down increment
-1

Load balancer

Load balancer visibility
public

Load balancer subnets
subnet-089884453914b91cd,subnet-02f232e411a24096f

Load balancer type
application

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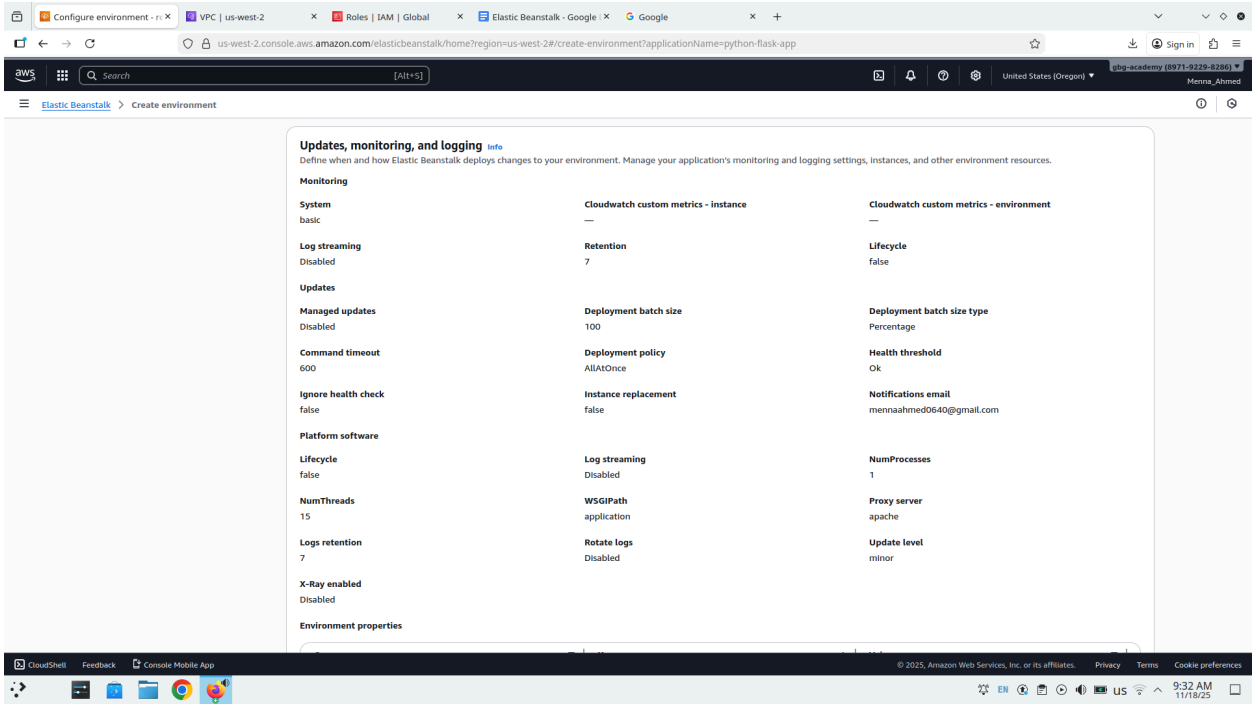

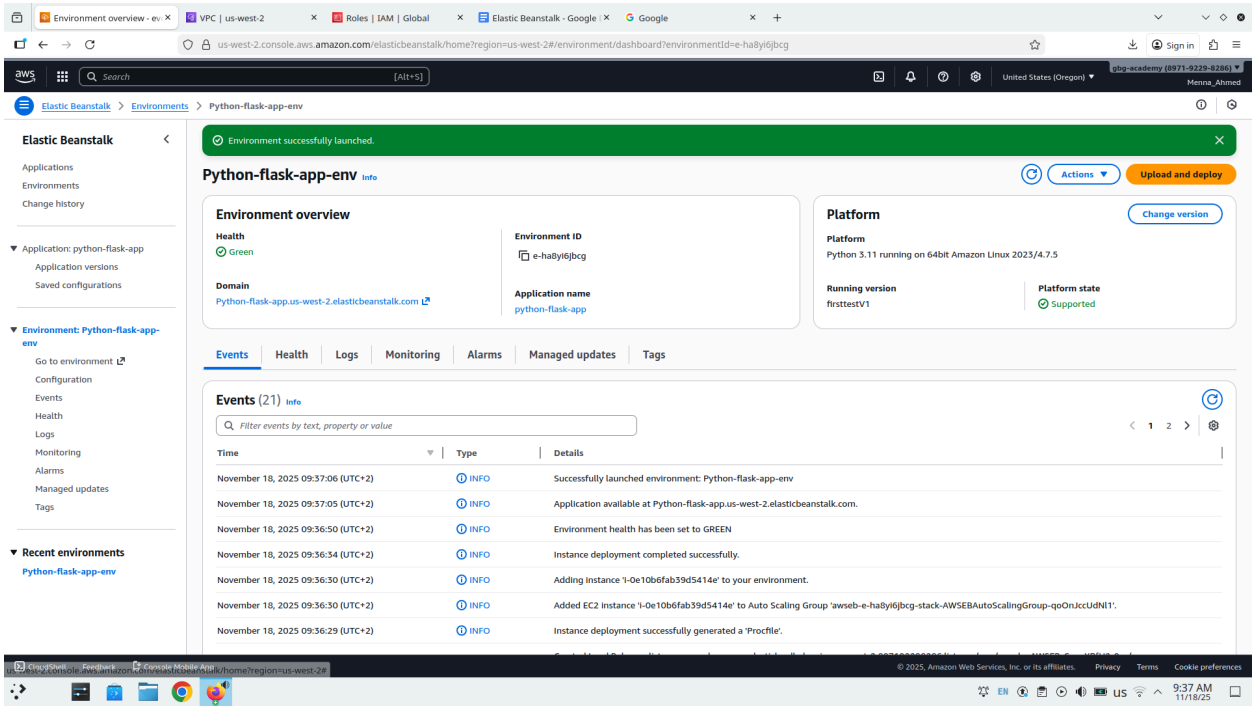

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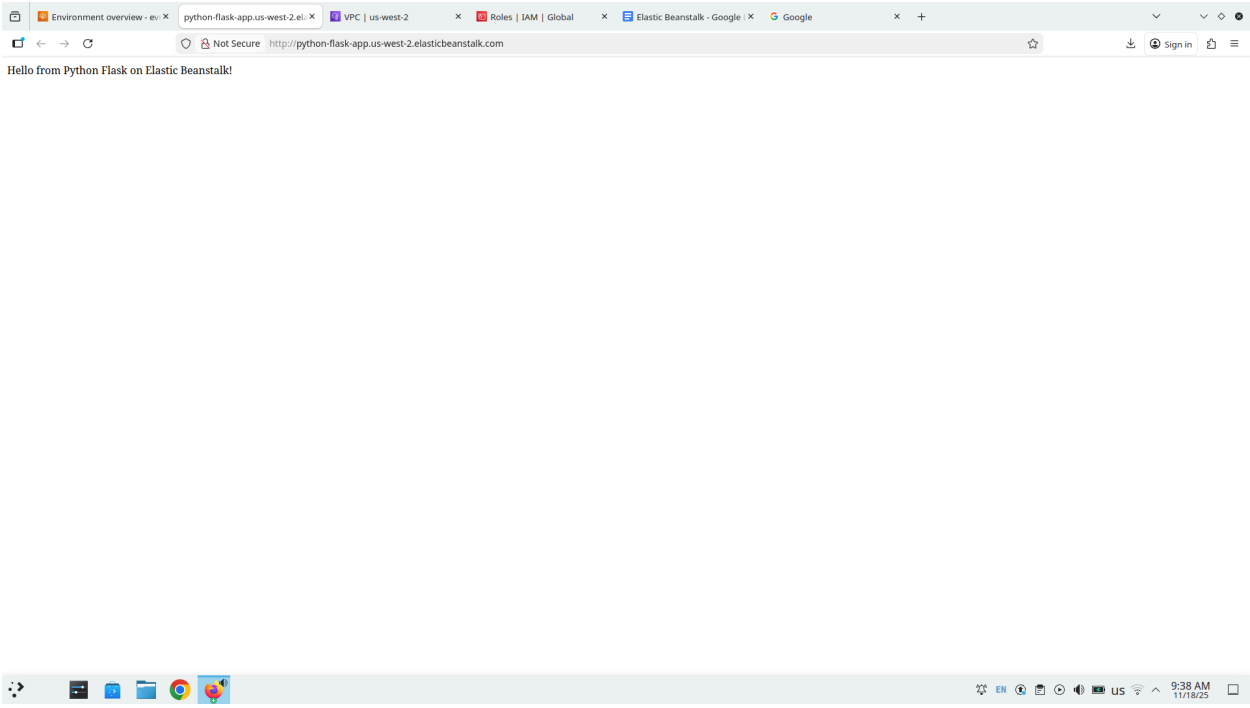

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